ZIANG TANG

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EDUCATION

University of Colorado Boulder — Boulder, CO, USA
Ph.D. Student in Computer Science — *Aug 2026 - Present*
Advisor: Prof. Mark Zhao

Rice University — Houston, TX, USA
Master of Computer Science — *Aug 2022 - May 2024*

Shanghai Jiao Tong University — Shanghai, China
Bachelor of Engineering in Information Engineering — *Sep 2018 - Jun 2022*

RESEARCH & WORK EXPERIENCE

Research Intern, Rice University — *Sep 2025 - Feb 2026*
Advisor: Prof. Jiarong Xing

- Research topic: Operator-level autoscaling for generative model serving, allocating resources at operator granularity to improve GPU, memory, and energy efficiency.
- Designed and implemented a modular operator-replication architecture on nano-vllm, enabling fine-grained scaling and parallel execution of model operators.
- Built an evaluation API server and benchmarked service-level objective (SLO) metrics, operator launch latency, and stream management overhead.

Software Engineer, Flexport — *Aug 2024 - Sep 2025*

- Developed full-stack logistics workflows for the Freight Forwarding App using React, Java/Spring, Ruby, and Scala.
- Implemented event-driven automation with AWS SQS, Kinesis, and Step Functions; developed gRPC APIs for interservice communication and managed relational data in PostgreSQL.

Research Assistant, Rice University — *May 2023 - Oct 2023*
Advisor: Prof. Akane Sano

- Research topic: Burnout Modeling and Analysis in Shift Workers Using Machine Learning.
- Developed machine learning models combining wearable and survey data to predict burnout risk in shift workers; used DiCE (Diverse Counterfactual Explanations) to identify feature changes that shift model predictions from high to low risk.
- Presented first-author research at IEEE BHI 2023 through a poster and rapid-fire talk.

OPEN SOURCE CONTRIBUTIONS

Contributor, kvcached — *Oct 2025 - Jun 2026*

- Implemented hybrid attention/Mamba support in vLLM and SGLang integrations, extending virtual-memory-backed cache allocation for mixed attention/SSM models (#310, #314, #318).
- Added GPT-OSS support across vLLM and SGLang and MLA support in vLLM, extending allocation across cache groups and unifying allocator interfaces (#259, #268, #261).

Contributor, vLLM — *Apr 2025 - Jul 2025*

- Implemented microbatch tokenization for concurrent requests with long prompts; merged into upstream vLLM (PR #19334).

PUBLICATIONS & PREPRINTS

OpScale: Operator-level Provisioning and Autoscaling for LLM Serving
Xingqi Cui, Chieh-Jan Mike Liang, **Ziang Tang**, Jiarong Xing, Haoran Qiu.
Revised manuscript under review; earlier version: *arXiv:2608.13499*, 2026.

vLLM Semantic Router: Signal Driven Decision Routing for Mixture-of-Modality Models
Xunzhuo Liu, Huamin Chen, Samzong Lu, et al. (including **Ziang Tang**). *arXiv:2603.04444*, 2026.

Burnout Prediction and Analysis in Shift Workers: Counterfactual Explanation Approach
Ziang Tang, Zachary King, Alicia Choto Segovia, et al. *IEEE EMBS BHI*, 2023.

HONORS & AWARDS

NSF Student Travel Award, IEEE EMBS BHI — *Aug 2023*
Outstanding Graduate Award, Shanghai Jiao Tong University — *Jun 2022*
Finalist, Mathematical Contest in Modeling (MCM) — *Mar 2020*
Annual Outstanding Undergraduate Scholarship, Shanghai Jiao Tong University — *2019 & 2020*